

Uber Supply Demand Gap

- From the analysis of 6,745 Uber trip requests, several operational challenges were identified. Peak demand occurs during the morning (5 AM–10 AM) and evening (5 PM–9 PM). Some number of requests remain unfulfilled due to No Cars Available, while cancellations also affect customer satisfaction. These insights were obtained through Excel dashboards, SQL queries, and Python EDA.
- A large percentage of ride requests could not be completed due to the unavailability of cars. This indicates that driver supply is insufficient during certain periods from Fig 1.1 .
- Uber should deploy additional drivers during
Morning: 5 AM – 10 AM
Evening: 5 PM – 9 PM
to reduce customer waiting time and improve trip completion rates from Fig 1.2 .
- Driver allocation should be adjusted dynamically between Airport and City based on demand to improve service efficiency from Fig 1.3 .

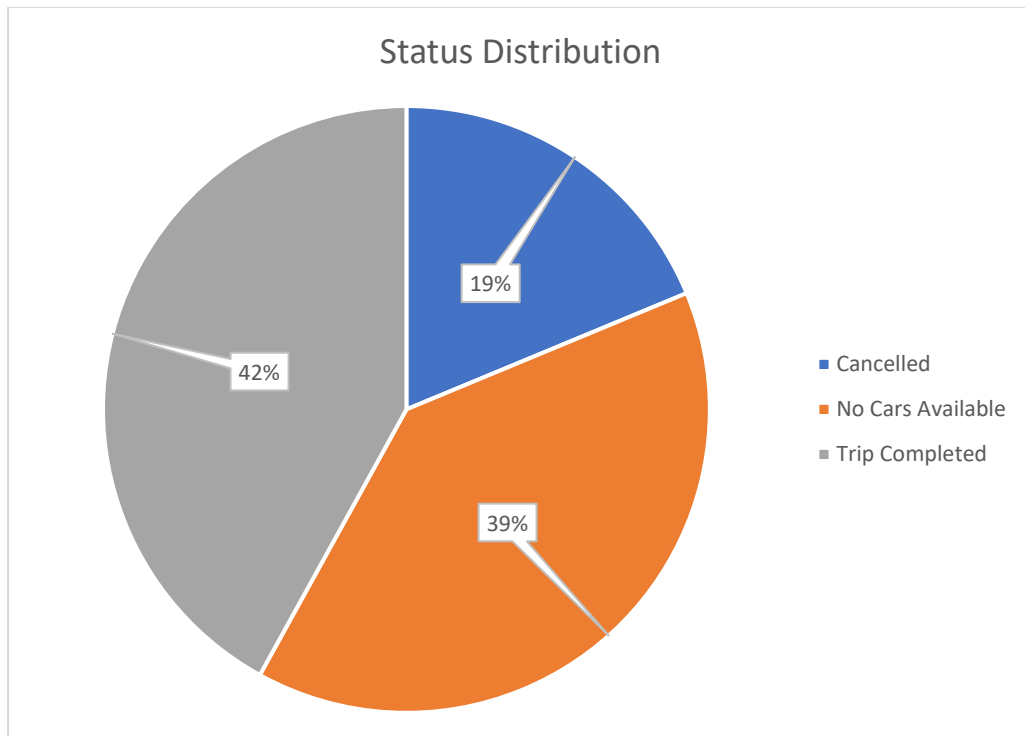


Fig : 1.1

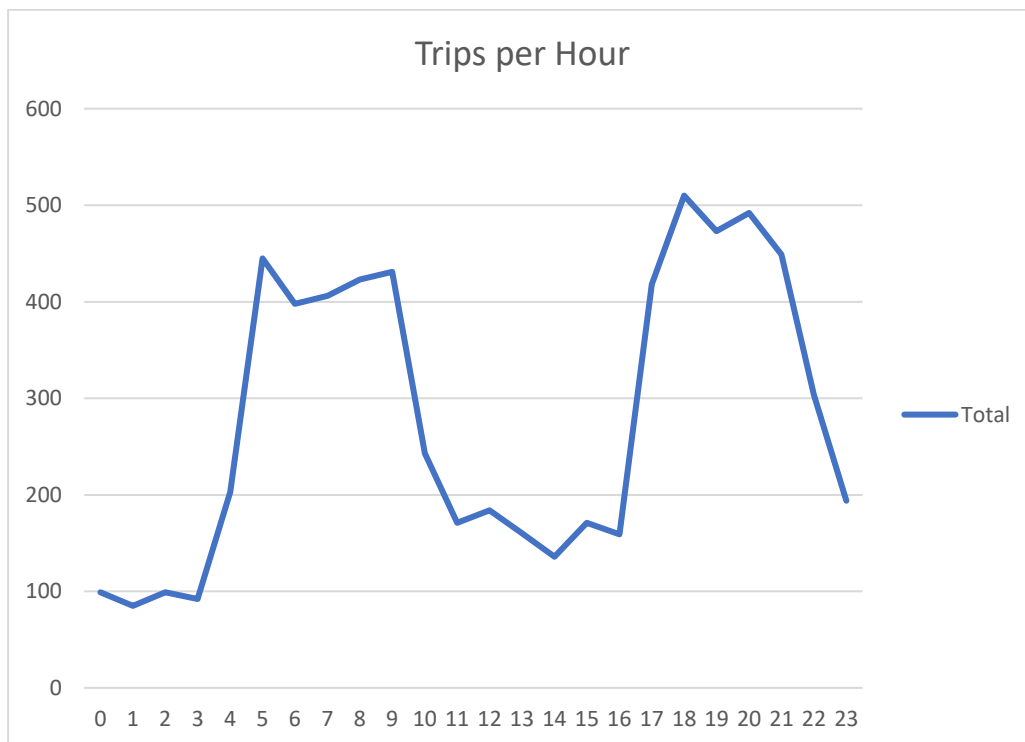


Fig : 1.2

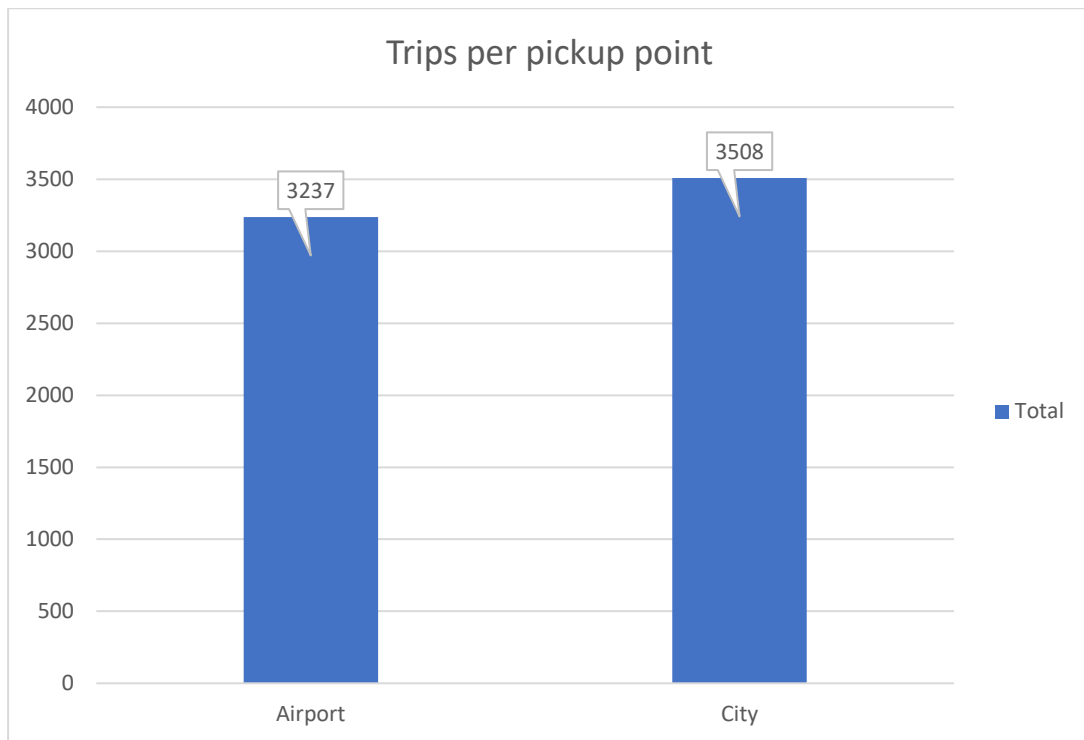


Fig: 1.3

SQL Insights

- The dataset contains a total of 6,745 trip requests.
- 42% of the requests were successfully completed, while 58% resulted in either cancellations or no cars being available.
- 39% of the requests could not be fulfilled due to No Cars Available, indicating a shortage of drivers during high-demand periods.
- 19% of the requests were cancelled, suggesting operational and driver availability issues.

- SQL analysis identified that morning (5 AM–10 AM) and evening (5 PM–9 PM) are the busiest hours for ride requests.
- Both Airport and City pickup points experience high demand, requiring better driver distribution.

Solutions for Supply Demand Gap :

- Increase driver availability during peak hours.
- Introduce incentive programs for drivers during high-demand periods.
- The SQL analysis highlights a significant gap between customer demand and driver availability during peak hours.
- Increasing the number of active drivers, optimizing driver allocation between pickup points can improve trip completion rates and reduce cancellations.
- Monitor trip completion, cancellations, and vehicle availability through dashboards.